

ETP3205: Innovation & Enterprise

Internship Report

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Executive Summary

As part of the NOC Vietnam program, I had the privilege of interning at Buymed, the largest e-commerce pharmaceutical distribution platform in Vietnam. Buymed, locally known as Thuocsi, operates as a B2B platform, delivering essential medical supplies and pharmaceuticals to clinics, hospitals, and pharmacies across the country. During my internship, I collaborated with both the Global Tech Team and AI Team, working on technical projects to enhance platform features, optimize data collection processes, and develop innovative AI solutions. Additionally, I gained valuable insights into the operational aspects of the startup, including fulfillment center logistics and supply chain management. This experience not only expanded my technical skills but also provided a comprehensive view of how a startup operates and scales in a competitive market.

As a software engineer intern in the Global Tech team, our initial responsibilities involved becoming familiar with Buymed's tech stack and working on the Thailand and Cambodia systems. For the first few weeks, Vaibhav, my other NOC intern, and I worked extensively to quickly learn their tech stack and develop new features for their global websites. Halfway into the program, we started working on a web scraping project to dynamically fetch competitor product data, enabling Buymed to make more informed pricing decisions. For the last two weeks of our internship, we collaborated with the AI team to research vector databases and develop a chatbot capable of answering queries about Buymed.

During my internship, I also had the opportunity to meet and learn from startup founders, developers, and entrepreneurship enthusiasts in Vietnam's startup ecosystem. We attended company bonding events, gatherings such as the SGN Connex Network event, and entrepreneurship events like the Start Up World Cup to learn about current startup ideas and network with professionals in the industry.

This internship not only greatly enhanced my technical skills and entrepreneurial knowledge but also contributed to significant personal growth. I became more open-minded in my interactions with others, embracing diverse perspectives and ideas. I also developed greater resilience and adaptability, learning to navigate and thrive in new environments and cultures. Although I faced challenges like a steep learning curve and communication barriers, I was able to overcome them with hard work and patience.

Company Background

Buymed, which translates to “medicine” in Vietnamese, is the largest ecommerce pharmaceutical distribution platform in Vietnam. Founded in 2018, Buymed has been rapidly expanding to meet the growing demand for pharmaceutical distribution across the region. The company's flagship product, Thuocsi.vn, is an online B2B platform that enables pharmacies, clinics, and hospitals to order a wide range of pharmaceutical products directly from manufacturers and distributors. This platform simplifies the procurement process, offering a vast catalog of products for customers and providing a reliable supply chain to healthcare providers. While based in Vietnam, the company has expanded its distribution services across Southeast Asia to countries such as Thailand, Cambodia, and Singapore, with a mission to solve defragmentation in healthcare.

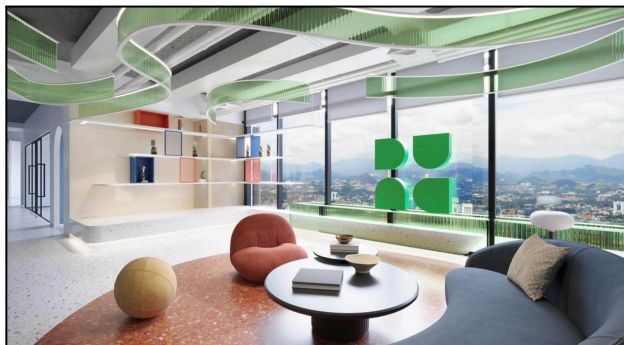


Figure 1: Buymed office

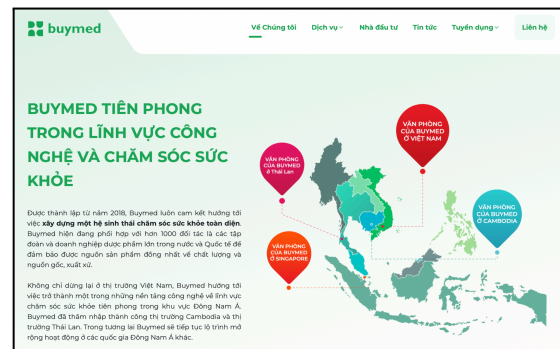


Figure 2: Buymed website

Buymed differentiates itself through its highly efficient supply chain process, which ensures that products reach customers quickly and reliably no matter where they are located. By partnering with reputable suppliers, Buymed maintains a steady flow of high-quality pharmaceutical products, directly sourced from trusted manufacturers. This strategic collaboration not only streamlines the procurement process but also enables Buymed to guarantee the authenticity of every product. Each order comes with detailed invoices and documentation, providing customers with the assurance that they are receiving legitimate, safe medications. This level of transparency and commitment to quality is an integral part of Buymed's operations, helping to build and maintain strong customer trust.

Buymed is led by a seasoned executive team, including founder and CEO Hoang Nguyen. With over 500 employees, and more than 70 members in the tech team, the company is organized into various major departments, such as technology, finance, marketing, and fulfillment & logistics team, each contributing to its rapid growth and success.

Since the beginning of my internship, Buymed has made remarkable strides across nearly all departments. In Cambodia, the company has reached significant milestones, notably surpassing 1 million GMV (Gross Merchandise Value) within a single month and achieving a 40% growth rate in June compared to May, with an additional 2% increase in July. The company has also recently integrated AI-driven tools to enhance various aspects of its operations, such as computer vision for inventory management and AI chatbots for customer support. Furthermore, Buymed has secured a strategic partnership with ABBank, paving the way for new opportunities within its ecosystem.



Figure 3: Partnership between Buymed and ABB Bank



Figure 4: Buymed logistics and fulfillment center

Before joining Buymed, I viewed the company as a fast-growing startup with a strong focus on health tech and an impressive B2B platform. However, my experience working there revealed that Buymed is not just a single entity but rather many companies combined: a technology firm, a supply chain organization, a pharmaceutical distributor, and soon, a financial services enabler. This multifaceted structure provided me with a rare opportunity to gain cross-industry knowledge and to understand the interconnected nature of various business domains. My time at Buymed has significantly deepened my understanding of the technical and business complexities inherent in pharmaceutical distribution and has enhanced my appreciation for how the company is driving change through its innovative approach and advanced technology.

Internship Details

During the early stages of our internship at Buymed, Vaibhav and I were assigned to the Global Tech team by our supervisor, Nam Pham. For the first two weeks, we familiarized ourselves with Buymed's tech stack, which included React, Next.js, and Nginx for the frontend, Go for the backend, MongoDB for the database, and Docker for deployment. We started with tutorials to familiarize ourselves with each language and application and demonstrated our learning by developing a personal notes web application. This hands-on project offered us a solid foundation in integrating frontend, backend, and database components effectively.

Once we became comfortable with the tech stack, we transitioned to working on the Thailand and Cambodia systems. We were assigned various tasks, or tickets, by our supervisor through Atlassian, a project management platform. These tasks ranged from simple bug fixes and UI modifications, such as translation updates and feature enhancements, to more complex features like video guide popups and referral link buttons. This experience allowed us to become well-acquainted with Buymed's codebase and development processes, fostering our ability to navigate and contribute to their complex technical environment effectively. Additionally, it helped us gain a deeper understanding of the company's workflows and the impact of our work on the overall user experience.

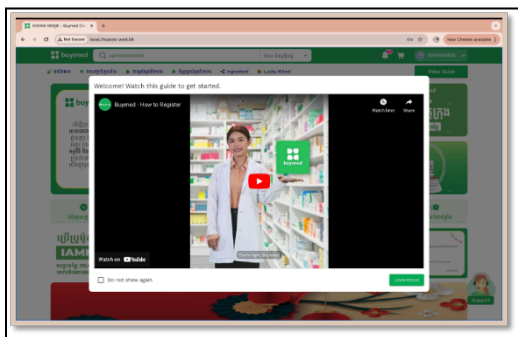


Figure 5: Video guide popup feature in Buymed Thailand system

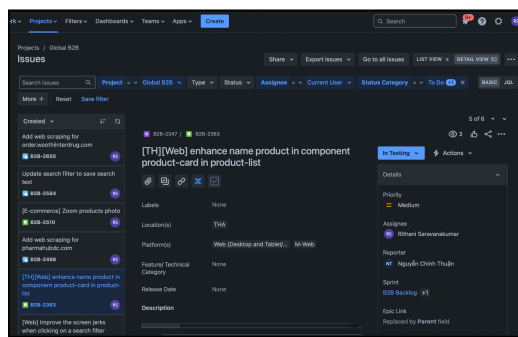
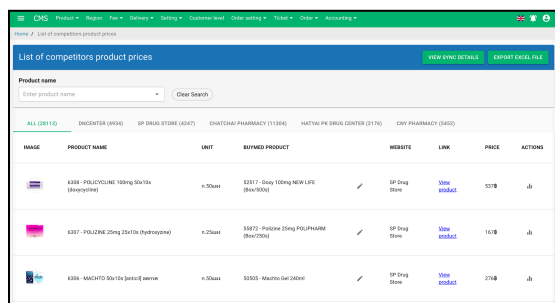


Figure 6: Atlassian ticketing platform

Approximately halfway through the program, we were tasked with the crawl price data project. The goal of this project was to scrape competitor product information from Thai pharmaceutical websites so Buymed could make more informed decisions regarding their product pricing. We were expected to fetch the product information, store it in Buymed's internal database, and integrate it into Buymed's internal system. We started by analyzing whether the websites used API calls for product data. Based on our findings, we developed API handlers to authenticate and fetch product information, either through direct API calls or by scraping HTML content. After thorough testing in Postman and coding in Go, we

deployed the crawlers on Buymed's servers. By the end, we managed to store product data from five different websites and ensured that the information was updated daily to reflect any changes made on the websites.

Following the backend development, we focused on presenting the data in Buymed's internal system. We added tabs for each website, implemented a search filter for easy navigation, and incorporated features like data export to Excel and crawler status monitoring. Additionally, with support from the AI team, we were able to use AI to map competitor products to Buymed's product catalog.



The screenshot shows a web application titled "List of competitors product prices". It features a search bar and a table with columns: IMAGE, PRODUCT NAME, UNIT, BUYMED PRODUCT, WEBSITE, LINK, PRICE, and ACTIONS. The table lists three products: 4388 - POLYCYCLINE 100mg 50x10 (Hydroxyline), 4387 - POLYLINE 50mg 50x10 (Hydroxyline), and 4386 - ANCHITO 50x10 (Jansin) amoxic.

IMAGE	PRODUCT NAME	UNIT	BUYMED PRODUCT	WEBSITE	LINK	PRICE	ACTIONS
	4388 - POLYCYCLINE 100mg 50x10 (Hydroxyline)	x.50mg	5217 - 100mg 100mg NEW LIFE (Biaxibio)	SP Drug Store	View product	5378	
	4387 - POLYLINE 50mg 50x10 (Hydroxyline)	x.25mg	5217 - 100mg 100mg NEW LIFE (Biaxibio)	SP Drug Store	View product	1078	
	4386 - ANCHITO 50x10 (Jansin) amoxic.	x.50mg	5335 - Maxtra Gel 240ml	SP Drug Store	View product	2768	

Figure 7: Buymed Internal CMS with competitor product information

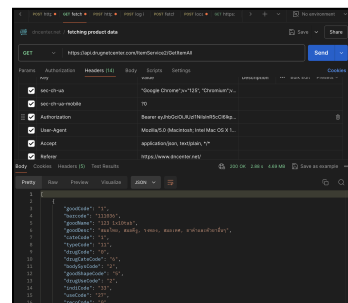


Figure 8: Testing API Handlers in Postman

In the final week and a half of our internship, we were transferred to the AI team to gain experience with AI tools. We began developing an AI chatbot designed to assist users in navigating Buymed's platforms. This involved researching vector databases and familiarizing ourselves with AI tools such as Langchain, OpenAI, Pinecone, and Streamlit. We used Python libraries to scrape data from Buymed's website, formatted the data using Pinecone, and trained a simple language model with OpenAI and Langchain.

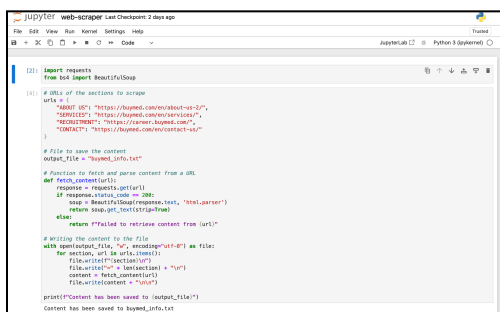


Figure 9: Code in Jupyter Notebook to scrape website

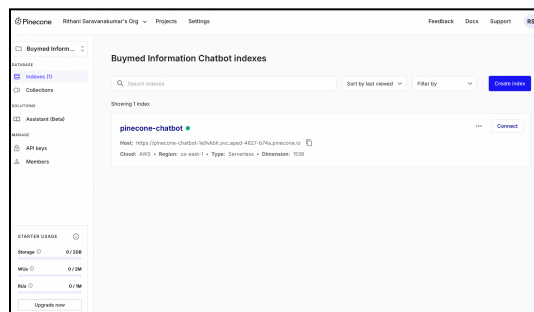


Figure 10: Pinecone vector database with chatbot index

Overall, this experience provided us with a comprehensive understanding of both technical and AI-driven projects at Buymed, contributing significantly to our professional development.

My role at Buymed evolved significantly over the course of the internship. Initially, I was primarily focused on software development and bug fixing. However, as I gained more experience and demonstrated my capabilities, I took on additional responsibilities such as working on the web scraping project and developing our own chatbot. Although these projects were demanding, they provided valuable opportunities for learning and growth, allowing me to expand my skills and make meaningful contributions to the team.

Through my involvement in these projects, I was able to learn about advanced software development techniques, industry coding standards, and deployment practices. The crawl data project, in particular, provided invaluable hands-on experience in building a comprehensive project from the ground up, encompassing frontend, backend, and database management. These experiences deepened my proficiency with Buymed's tech stack, specifically Go, Nginx, and Postman, and broadened my familiarity with a diverse range of technologies. Additionally, I was able to get exposure working with AI tools such as Langchain, Pinecone, and OpenAI for the very first time.

One of the primary challenges I encountered was the steep learning curve across all the projects, as I had to rapidly become proficient with a wide range of new tools and technologies. From learning about frontend languages, to backend tools, to different databases, and AI technologies, I sometimes found it overwhelming to keep up with the team's expectations. Additionally at times, it was hard to communicate with our colleagues, which resulted in difficulties with understanding project requirements. However, I was able to overcome these challenges by proactively seeking out resources, dedicating extra time to learning, and leveraging the support of my colleagues and mentors.

In hindsight, I realize it would have been advantageous if we had been transitioned to the AI team much earlier, with a few weeks remaining in the internship. This would have provided us with more time to deeply understand the tools and contribute more effectively to the chatbot development. Moreover, having grown close to our global team colleagues, it was difficult to part ways close to the end of the internship.

Aspects of Personal Growth

Living in Vietnam was a huge turning point for my personal growth. It was the first time I had lived alone in a completely new environment, which meant I had to quickly learn how to navigate not just the physical space but also the cultural nuances. I learned how important it is to be direct and decisive in my interactions. I started being more open and proactive, asking for what I needed without second-guessing myself, whether it was a simple recommendation or help with something at work. This shift made me less caught up in overthinking and more confident in my decisions, allowing me to adapt much quicker. I also made it a point to push myself out of my comfort zone by striking up conversations at meetups and networking events like the SGN Connex Network Event and the HOC HCM Alumni Event. This approach not only boosted my resilience but also helped me handle the challenges I faced in Vietnam with a more positive and flexible mindset.

During my time at Buymed, I picked up valuable business skills that broadened my understanding of the startup world. I learned about market adaptation and strategic pivoting, especially after listening to the CEO's stories about his health tech startup in the US. These experiences showed me just how important it is to be agile and responsive to market changes, and how crucial it is to pivot strategies based on feedback to stay competitive. Similarly, while working on the crawl data project, I was able to get a strong grasp of market analysis. By gathering and analyzing competitor product info, I helped Buymed spot market trends and opportunities, which directly shaped our product strategy. This data-driven approach gave me a solid understanding of how market insights can steer a company's decisions.

I also gained a deeper understanding of Buymed's operational challenges, especially during the COVID-19 pandemic. It became clear how much external factors can affect business operations and why it's so important to adapt strategies to keep things stable and growing during tough times. Lastly, I learned about the logistics and processes involved in Buymed's fulfillment center, like managing invoices and processing orders. This gave me a clear picture of how operational efficiencies are achieved and how important software systems and AI are in making these processes run smoothly.

Entrepreneurship Related Activities

Throughout my time in Vietnam, I actively participated in a range of entrepreneurship-related activities that have significantly contributed to achieving my goals. These activities played a crucial role in enhancing my understanding of the startup ecosystem and provided valuable learning experiences.

One of the key events I attended was the SGN Connex Network Event. This event brought together students from Singapore universities who were currently living or working in Ho Chi Minh City and offered a comfortable environment for networking and knowledge exchange. It was particularly inspiring to connect with fellow Singaporean students and hear about their startup experiences and entrepreneurial ventures in Vietnam. Learning about the diverse projects they were working on and the challenges they faced in a new market was eye-opening. These interactions not only broadened my perspective on entrepreneurship but also reinforced the value of community and collaboration among young entrepreneurs navigating similar paths in a foreign environment.

Another significant event was the NOC HCM Alumni Event. This gathering provided a unique opportunity to connect with the current NOC participants, some of whom were working on their entrepreneurial ideas, and NOC alumni. The alumni's stories of navigating the startup journey, from ideation to execution, were both inspiring and educational. I learned about the importance of resilience and adaptability in the face of setbacks and the value of a strong support network. These conversations helped me appreciate the importance of persistence and the need for continuous learning in the entrepreneurial process.

In my opinion, the Startup World Cup was the highlight of my entrepreneurial experience. This event featured pitches from some of the most promising startups in the region, offering a firsthand look at the innovation driving the startup ecosystem. I observed how founders effectively communicated their value propositions and addressed potential investors' concerns. This exposure taught me the importance of crafting a compelling pitch and being able to articulate a clear business strategy. I also learned about different funding strategies and what investors prioritize when evaluating startups, which will be invaluable as I pursue my own entrepreneurial endeavors.

In addition to these external events, participating in company bonding events and interacting with various teams at Buymed gave me a deeper understanding of cross-functional collaboration and team dynamics within a startup. These interactions were crucial in refining my ability to work effectively in diverse teams and understand different aspects of business operations, such as marketing, product development, and

customer service. I gained practical insights into how different departments collaborate to drive the overall success of the company.

Reflecting on these experiences, I feel they have been aligned with the goals I set at the beginning of the program. They have helped me build a solid foundation in entrepreneurship and provided me with practical insights into running a startup. However, if I were to do anything differently, I would consider attending a wider variety of activities, including more industry-specific workshops and seminars, such as AI and technopreneurship related events. This would offer deeper insights into specific areas of entrepreneurship and help me acquire targeted skills relevant to my interests in technology and healthcare.



Figure 11: SGN Connex Network Event



Figure 12: NOC HCM Alumni Networking Event



Figure 13: Startup World Cup event on 21 July 2024

Entrepreneurship Focus and Outcomes

Having been immersed in Buymed's dynamic and entrepreneurial environment for the past three months, I was able to explore several innovative startup ideas. One particular concept that came to my mind was the development of a comprehensive competitor analysis tool designed specifically for the e-commerce sector. Inspired by our crawl price data project and chatbot, this tool would leverage web scraping and data analytics to provide real-time insights into competitor pricing, product offerings, and market trends. By integrating AI and machine learning algorithms, the tool could offer predictive analytics and actionable recommendations to help businesses adjust their strategies proactively.

To pursue this idea, I plan to leverage the skills and knowledge I gained during my internship, particularly from the crawl data project and chatbot, to build a prototype. My goal is to develop a minimum viable product (MVP) and conduct testing with potential users to refine its features and functionality. As a member of the NUS Entrepreneurship Society's executive committee, I also intend to share this idea with my peers and recruit interested individuals to collaborate on bringing this project to life.

In terms of facilitating exploration and experimentation during the program, I think incorporating structured innovation workshops and hackathons could be highly beneficial. These activities would encourage interns to brainstorm and prototype ideas in a collaborative environment, fostering creativity and practical problem-solving skills. Apart from the work I did at my company, I felt like there were no additional opportunities to showcase my knowledge and creativity. I believe partnering with local organizations that offer these programs for students would be very helpful for those doing NOC in the future.

Regarding the program's assessment methodology, I found it to be quite effective in holistically evaluating the student's experience. The use of three different forms—the logbook, report, and presentation—provides a well-rounded approach to documenting and reflecting on each intern's learning journey. Additionally, the check-in at the midpoint of the program was a valuable touch, offering a chance to gauge progress and adjust our goals if needed. I think it would be even more beneficial if the program included a group discussion with all NOC students as part of the participation grade. This would not only encourage more frequent interactions but also foster a stronger sense of community.

Overall, my time in Vietnam has been nothing short of exceptional - I have been able to learn tremendously from my colleagues at work and my peers on the program, all while getting to network with skilled entrepreneurs and travel the country. The experiences I have gained from this NOC program are truly invaluable, and I know that I have made memories that I will cherish for a lifetime.